CS5013 Project Proposal

Unified Lab Project & Idea Tracking Platform

Priyal Patel, CS24B047 Sujal Jayant Kumar, DA24B026

August 21, 2026

1 Team information

Team ID: team-47

- Member 1: Priyal Patel – roll no.: CS24B047 – department: CSE – GitHub handle: P31P
- Member 2: Sujal Jayant Kumar – roll no.: DA24B026 – department: DSAI – GitHub handle: SujalKumar06

Primary contact email: cs24b047@smail.iitm.ac.in

2 Problem statement

There is a need for a unified platform for keeping a record of which projects are active, where they stand, which ideas students have come up with or which papers have been shared within the lab.

Project and idea tracking. The existing tracker requires multiple mandatory fields behind a slow login and refreshing, so updates are often not submitted and early-stage ideas are usually not recorded at all.

Paper sharing. Papers are circulated as Slack links and stored nowhere else, so retrieving an earlier one means scrolling the channel history.

3 Users and stakeholder

Named stakeholder: Raman Labs (Head: Prof. Karthik Raman), Department of Data Science and AI, IIT Madras.

Primary point of contact: Sabdhayini K B, Research Student, Raman Lab.

We first reached out to Prof. Karthik Raman on 12 August 2026, who directed us to work with Sabdhayini to understand the lab's daily workflow. Sabdhayini detailed the lab's needs and what implementations can help the lab function better. She continues to assist us by giving her inputs on behalf of the lab and reviewing our project drafts.

The stakeholder's acknowledgement email is attached as Appendix A.

4 Prior work and why it does not suffice

1. **Similar Website for Project tracking** – https://ramanlab-project-tracker-5eqj.onrender.com/.

Overlap: Tracks projects and ideas in a similar fashion as suggested.

Gap: Every submission demands a long list of mandatory fields, so members stop filing updates and the data goes stale. It is deployed on Render's free tier, which cold-starts after a few idle minutes and takes tens of seconds to respond. There is no connection to Slack or to the papers the work is based on.

2. **Manual Workflow (Excel Sheets & Chat Logs)** – Internal / manual.

   Overlap: For Slack, the links of research papers need to be recorded manually. For Whatsapp there is an excel sheet that tracks them.

   Gap: As there is no integration with slack, paper links get buried in messages. With the excel too, they have to manually add the links from Whatsapp here.

3. **Scite** – https://scite.ai

   Overlap: Extracts references and builds citation context for a given paper.

   Gap: Paid, with a restrictive free tier, and it solves only the citation half. It has no notion of the lab's projects or ideas, and no Slack integration, so a member would still have to move every link across by hand.

5 Tech stack

Language: **Java** (course policy).

- Build system: Maven
- Framework: Spring Boot
- Key libraries: Slack Bolt for Java (Slack Bot integration), Jackson (JSON handling), Java HttpClient (LLM API calls for Auto-Fill), PostgreSQL Driver also Javascript for Frontend.
- Where the tool will run: IIT Madras department server / hosted VM

6 AI-usage statement

We will use AI coding assistants (such as Claude Code, GitHub Copilot, ChatGPT, or Gemini) during this project, in line with the course policy on AI-assisted programming. Every line of code we submit will be code we understand and can explain in the viva, whether we typed it or the assistant generated it.

7 Verification plan

Test fixtures we will use:

- An export of historical Slack channel messages from Raman Lab containing diverse paper link formats (e.g., arXiv URLs, DOIs, direct PDF links).
- Synthetic unstructured text inputs (including short fragments, missing fields, and special characters) to test the LLM auto-fill feature.
- A mock dataset of 15–20 ongoing lab projects with varying timelines, status tags, and update histories to verify database constraints and UI rendering.

User-visible acceptance criteria the stakeholder will use to say "yes, this works":

- Automatically detects research paper links shared in the Slack channel and archives them along with link in the web platform without manual input.

- Makes the user experience for form filling etc seamless without glitches or high wait times.

8 Milestone plan

- **By the design doc (11 Sep 2026):** Set up user login, create the core database structure, and establish an initial connection route with Slack.
- **By the mid-demo (9 Oct 2026):** Show a working web interface running locally with visual project timelines, simple idea logging, and automatic saving of research-paper links and their references from Slack.
- **By the final submission (6 Nov 2026):** Add the AI auto-fill feature for new ideas, clean up the visual design, and host the app on the department server so the lab can run it. Further Scope -look into speech to text to make the form-filling more seamless.

9 Risks and plan B

1. **Risk:** Slack Webhook does not work due to any access/approval issues, so it doesn't poll messages to get the links.
   Plan B: Set up a cron job that polls messages every few minutes and gets the links.
2. **Risk:** We don't get a servrer to host the web app due to access issues etc.
   Plan B: Look into private domian names, third party hosting services that don't have much refresh time.
3. **Risk:** Not comfortable sending the information to a LLM.
   Plan B: Use local open source LLMs.
4. **Risk:** If we could not scrape the references of the Research Paper.
   Plan B: Just mention it to the user so that it doesn't silently fail and leave out info from the user.

A Stakeholder acknowledgement email

The following acknowledgement email was received from Prof. Karthik Raman, Head of Raman Labs, Department of Data Science and AI, IIT Madras, on 21 August 2026.

Project + Idea Tracker Proposal for Acknowledgement

2 messages

Sujal Jayant Kumar da24b026 <da24b026@smail.iitm.ac.in> Fri, 21 Aug, 2026 at 9:03 am
To: Karthik Raman <kraman@dsai.iitm.ac.in>
Cc: cs24b047@smail.iitm.ac.in

Respected sir,
After talking to Sabdhayini from your lab and continuous discussion about the project that we informed you about, we would like to go forward with the project and we have come up with the following proposal (PFA) to submit for our course.
We would be in touch with you and Sabdhayini so we can continuously get feedback.

Requesting you to please acknowledge on behalf of Raman Labs so we can proceed further.

Thanks and Regards

Sujal Jayant Kumar, DA24B026
Priyal Patel, CS24B047

project_proposal.pdf
146 KB

Karthik Raman <kraman@dsai.iitm.ac.in> Fri, 21 Aug, 2026 at 9:24 am
To: Sujal Jayant Kumar da24b026 <da24b026@smail.iitm.ac.in>
Cc: cs24b047@smail.iitm.ac.in

Super!!! Looking forward to see how this shapes up! Good luck to you & Priyal.

Pls. consider the proposal acknowledged on behalf of Raman Lab. You may proceed with the course submission and continue coordinating with Sabdhayini for feedback.

Regards,
Karthik

[Quoted text hidden]